1. Identity

Document	STAR-PKG-STAR-MANEUVER-EN · v2.1 · 2026-11-01
Package	star-maneuver
Version	0.2.2
Capability	star.orbit.maneuver
Maturity	VERIFIED
Python	⩾3.9
Source commit	94fcf7c
Source	https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_maneuver

2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE file and the copyright notices (and a NOTICE file, if a future version adds one), and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_maneuver-0.2.2-py3-none-any.whl
```

FILE	SHA-256
star_maneuver-0.2.2-py3-none-any.whl	88adff83800c1922219ac776d63e34abb0397042121e6a476a6d122dea53c476
star_maneuver.py	c1908a47917dfdba2b61883b2d442ec547e51c6bc4986e94bb71d4ef7d8dfc83
star_maneuver-0.2.2-source.zip	931f9923b161227bda7d5a4f6e412c20e41065f2c70ea2aa843591fd401299d1

5. What it does, and its limits

Impulsive manoeuvre arithmetic between circular coplanar orbits and plane changes. Standard library only.

Finite burns, gravity losses, non-coplanar Hohmann with optimal plane-change split, any flight qualification.

- `vis_viva(r, a, mu)`; `hohmann(r1, r2, mu)` and `bielliptic(r1, r2, rb, mu)` return per-burn and total delta-v (km/s) and time of flight (s). Units: km, s, km/s, km^3/s^2 (default mu = 398600.4418, IERS Conventions 2010).
- `plane_change(v, di)` = 2 v sin(di/2); `combined_dv(v1, v2, di)` by the law of cosines (radians).
- Invalid input fails explicitly with `ValueError` (non-positive radii, rb < max(r1, r2), negative speed).

6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

7. Citation and bill of materials

```
@software{cavazzini_star_maneuver_0_2_2,
  author  = {Cavazzini, Andrea},
  title   = {{star-maneuver}: Closed-form vis-viva and Hohmann orbital transfer engine},
```

```
  version = {0.2.2},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_maneuver/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

8. Warranty and liability

Provided "as is", without warranties or conditions of any kind (Apache License 2.0, sections 7 and 8), to the extent the law allows. This never excludes liability that the law does not allow to exclude.

Related: space activities and export control · acceptable use · evidence and limits · licence texts